Student Startup Crowdfunding Platform

UML Component Diagram — Layered Architecture

PRESENTATION LAYER

«component»
Web Client UI
Founder, Backer and Faculty Committee screens (browser, touch or mouse)

BUSINESS LAYER

ICampaign
REST / HTTPS (JSON)
createCampaign, selectTier, pledge

IMilestone
REST / HTTPS (JSON)
submitEvidence, requestTranche, decide

«component»
Campaign Manager

- Create campaign, define milestones (FR-003)
- Reward-tier selection (FR-002)
- Pledge orchestration, tier validation
- Early-bird bonus (48 h window)
- Owns campaign and pledge state

«external system»
Payment Gateway
PCI-DSS compliant, token-based

IGateway
HTTPS, tokenized request
authorize / capture / transfer

IPayment
internal API call
authorizePledge(token)

«component»
Payment Service
Tokenized authorization, escrow ledger and tranche disbursement (NFR-001)

IPayment
internal API call
releaseTranche(id)

«component»
Milestone Review Service

- Accept milestone evidence (FR-004)
- Faculty Committee review and decision record (NFR-002)
- Tranche release on approval (FR-001)
- Triggers backer notification
- Blocks release without evidence

INotify
async message
notifyBackers(tranche)

«component»
Notification Service
Email / in-app alerts to every backer of the campaign (FR-005)

DATA LAYER

IDataAccess
SQL / JDBC
campaign, tier, pledge records

IDataAccess
SQL / JDBC
escrow ledger entries

IDataAccess
SQL / JDBC
milestone, evidence, decisions

«component»
Database Component
Campaigns, reward tiers, pledges, escrow ledger, milestones, evidence, committee decisions

LEGEND

Component · External system · Provided interface (ball) · Required interface (socket) · Port · Direction of call (requirer → provider)

Each ball-in-socket pair is an assembly connector. Interface labels give the protocol and the operations passed.